

Öcal Kaptan

APPLIED AI ENGINEER & WORKFLOW AUTOMATION

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Residence Permit Status: **Permanent Residency (Germany)**

PROFESSIONAL EXPERIENCE

Mercor (Remote), Freelance: Applied AI & Automation

since June 2025

- Designed and shipped multi-stage n8n workflows orchestrating Claude with Airtable as the data layer and webhooks for integration; reduced manual content effort by 70%.
- Built prompt libraries and reusable structured-output instructions; ran iterative evaluations comparing output quality, latency, and cost to select the right model and prompt per step.
- Built a webhook-driven automation agent integrating third-party APIs (Meta, YouTube, Cloudinary) with human-in-the-loop review; deployed and monitored the pipelines on Python, FastAPI, and AWS with Git version control and clear documentation.

University of Duisburg-Essen, Research Assistant

Apr. 2023 – Apr. 2025

- Conducted code reviews and API-based data preparation for the open-source textbook *Econometrics with R*, used by thousands of students worldwide.
- Structured and automated research data from 10+ parallel projects; wrote reusable scripts and documentation so non-technical researchers could reproduce results.

TELUS International AI, Data Analyst (Working Student)

Sep. 2020 – May 2021

- Analyzed 50,000+ datasets, performed anomaly detection, and built KPI dashboards; reduced policy violations by 30%.
- Worked under strict data-privacy and compliance requirements with access controls and audit trails.

Çelebi Aviation Holding, Flight Operations Officer

Feb. 2014 – Jul. 2019

- Coordinated 20–25 parallel flights daily with standardized handovers and clear escalation under real-time pressure.

PROJECTS (SELECTED)

AI Automation Workflows for E-Commerce, n8n, Claude API, Airtable, Webhooks, JavaScript

2026 – ongoing

- Open-source n8n workflows orchestrating Claude and Airtable to generate e-commerce content at scale (product SEO, descriptions, support-ticket triage) for an online-retailer use case.
- Human-in-the-loop by default, structured JSON outputs with fence-stripping for safe parsing, rate-limit-aware batch dispatch, and a cost-driven model choice (Claude Haiku).

GridOps Energy Forecasting Pipeline, Python, XGBoost, FastAPI, AWS

Dec. 2025

- End-to-end ML pipeline with automatic model selection and a live SHAP explainability dashboard; achieved 2.19% MAPE and 1.8 GW RMSE.

Rental Data Pipeline & Analytics, Python, PostgreSQL, AWS

Mar. 2023

- Built an ETL pipeline for 50,000+ records into a normalized PostgreSQL schema; developed 15+ REST API endpoints for analytics.



CORE STRENGTHS

AI Tooling & Foundation Models:

Daily hands-on use of LLMs (primarily Claude) for prototyping and production. Practical understanding of multimodal and image/video generation models and how to choose between them.

Workflow Automation:

Build AI-orchestrated workflows with n8n and webhooks, connecting model steps to data sources and human-in-the-loop review.

Prompt & Context Engineering:

Draft, iterate, and document reusable prompts, structured-output patterns, and basic RAG / embedding pipelines.

Evaluation & KPIs:

Maintain golden test sets; benchmark models on quality, latency, cost, and hours saved. Comfortable with KPI and cost dashboards.

Stakeholder Translation:

Break non-technical pain points into user stories and scoped technical specs, then document for reuse.

TECHNICAL SKILLS

Languages: Python, SQL, R, Shell, JS/TS (reading)

AI & Automation: Claude / LLMs, n8n, RAG & embeddings, prompt engineering, structured outputs

Integration: REST APIs, Webhooks, FastAPI, third-party APIs (Meta, YouTube, Cloudinary)

Data & Pipelines: Apache Spark, dbt, Airflow, PostgreSQL, ETL

Platforms & Cloud: AWS (GCP concepts), Docker, Git/GitLab CI/CD, Airtable, Microsoft 365

EDUCATION

M.Sc. Econometrics – TU Dortmund

Thesis: Demand Forecasting with Neural Networks & SHAP 2021–2024

B.Sc. Econometrics – Karadeniz Technical Univ.

Thesis: Machine Learning for Retail Forecasting 2010–2014

LANGUAGES

English (C2) · German (B2) · Turkish (Native) · Spanish (A2)